



M-Tech Production & Marketing

Technology | Training | Marketing | Projects

Week 3 - Day 5 Report

Internship Program — Batch 2026

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1. Introduction

This report presents the work completed for the Neural Networks experiment during the MTech Production & Marketing AI/ML Internship Program 2026. The experiment introduced the fundamental concepts of Artificial Neural Networks (ANNs), which form the backbone of Deep Learning. It explained how interconnected artificial neurons process information and learn from data to solve complex prediction and classification problems.

2. Objectives

- Understand the fundamentals of Artificial Neural Networks.
 - Learn the architecture of input, hidden, and output layers.
 - Study the concepts of forward propagation and backpropagation.
 - Understand the role of weights, biases, and activation functions.
 - Explore practical applications of Neural Networks.
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3. Task Description

The objective of this experiment was to study the internal structure and learning process of Artificial Neural Networks. The experiment explained how data moves through different layers, how predictions are generated, and how errors are minimized using backpropagation. It also introduced commonly used activation functions and optimization techniques used in deep learning models.

4. Work Done

During this experiment, I studied the architecture of Artificial Neural Networks and learned how individual neurons work together to process input data. I understood the purpose of weights, biases, and activation functions in producing meaningful outputs from the network. I explored the concepts of forward propagation, where input data flows through the network to generate predictions, and backpropagation, where prediction errors are used to update model parameters. I also learned about commonly used activation functions including ReLU, Sigmoid, and Softmax, and understood how they improve the learning capability of neural networks. In addition, I reviewed the practical applications of neural networks in image recognition, speech processing, recommendation systems, healthcare, and natural language processing.

5. Challenges Faced

- Understanding how backpropagation updates network parameters.
- Learning the mathematical relationship between weights, biases, and gradients.
- Understanding the purpose of different activation functions.

6. Key Learnings

- Artificial Neural Networks imitate the learning process of the human brain.
- Neural networks consist of multiple interconnected processing layers.
- Backpropagation helps reduce prediction errors during training.
- Activation functions enable neural networks to solve complex problems.
- Neural Networks are widely applied in computer vision, speech recognition, NLP, robotics, and healthcare.

7. Conclusion

This experiment provided valuable practical and theoretical knowledge about Artificial Neural Networks and Deep Learning. I learned how neural networks process information, improve their performance through training, and solve complex AI problems. The concepts covered in this experiment will help me understand more advanced deep learning models and real-world AI applications.

Intern Signature:



Supervisor:

Name: Ghazi Muhammad Abdullah

Name:

Date: 17 July 2026

Designation: